Pivotal-based Classical and Bayesian Estimation for Block Generalized Adaptive Progressive Hybrid Censoring for the Kumaraswamy Distribution

Shreyas Samantray & Tankasala Madhughnesh
Mehta Family School of Data Science and AI, IIT Guwahati, India

Abstract

Censoring is frequently encountered in reliability and life-testing experiments where observing the lifetimes of all experimental units may be impractical due to time and resource constraints. This study considers classical and Bayesian estimation for the Kumaraswamy distribution under a block generalized adaptive progressive hybrid censoring framework with a common shape parameter and block-specific parameters. The block structure allows lifetime information from multiple groups to be combined, while the censoring scheme accommodates restrictions on the duration and progress of the experiment. We develop unrestricted and order-restricted maximum likelihood estimation, Fisher-information-based and per-block pivotal confidence intervals, and Bayesian estimation under independent Gamma priors, both without order restrictions and subject to an ordering of the block-specific parameters. The procedures are compared in a simulation study and applied to insulating-fluid breakdown data. In the simulations the order restriction reduced the mean squared error of the lower block parameter when the block parameters were close, the Fisher and credible intervals had near-nominal coverage, and the Bayes estimators were sensitive to the prior. The per-block pivotal intervals are approximate, rather than exact, under the adaptive scheme, and their coverage is assessed empirically.

Keywords: Kumaraswamy distribution; block censoring; generalized adaptive progressive hybrid censoring; maximum likelihood estimation; Bayesian estimation; order restrictions; pivotal inference.

1 Introduction

In reliability analysis and life-testing experiments, the assessment of product lifetimes is an important problem. In many practical situations it is not feasible to observe the failure times of all experimental units because of limitations on testing time, cost, and other experimental resources. Consequently, some observations may be censored, and appropriate statistical methods are required to make efficient use of the available lifetime data. Progressive censoring schemes provide flexibility by allowing surviving units to be removed from an experiment at specified failure times. Hybrid censoring schemes combine restrictions based on the number of observed failures and the duration of an experiment. Generalized adaptive progressive hybrid censoring (GAPHC) extends this framework by adapting the removal pattern to the progress of the experiment while guaranteeing that the test ends by a pre-assigned time. Lee and Lee [1] studied likelihood inference under a generalized adaptive progressive hybrid censoring scheme for the exponential distribution. Such designs are relevant when an experimenter needs to balance the amount of failure-time information collected against the time and resources required to complete the experiment.

In applications involving multiple groups, facilities, or production lines, the experimental units may be organized into blocks. A block censoring framework can account for differences between groups while representing the structure of the experiment. Related work has investigated

block progressive censoring for inverted exponentiated models [2] and block hybrid-censored Weibull populations [3]. Combining block structure with generalized adaptive progressive hybrid censoring provides a framework for studying lifetime data collected from multiple groups under practical experimental constraints.

The choice of an appropriate probability distribution is another important aspect of reliability analysis. The Kumaraswamy distribution is a two-parameter continuous distribution supported on the interval $(0, 1)$. Its cumulative distribution function and probability density function have simple closed forms, and the distribution can represent different shapes for suitable parameter values. Kumaraswamy models have been studied under generalized progressive hybrid censoring, including in work on partially observed competing risks [4]. Reliability estimation for the Kumaraswamy distribution has also been considered under block progressive Type-II censoring [5]. These studies provide relevant background for investigating Kumaraswamy models under more structured censoring schemes.

Parameter estimation under censoring can be challenging because the observed data contain incomplete information about the lifetimes. Maximum likelihood estimation provides a classical approach to estimating unknown parameters from the observed failure and censoring information. Bayesian estimation offers an alternative by combining the likelihood with prior distributions representing available information about the parameters. In addition, when the scientific context supports an ordering among block-specific parameters, order-restricted estimation can incorporate that information into both classical and Bayesian procedures. The usefulness of such restrictions depends on whether the assumed ordering is appropriate for the application.

Motivated by these considerations, this study investigates classical and Bayesian estimation for the Kumaraswamy distribution under a block generalized adaptive progressive hybrid censoring framework. We consider unrestricted and order-restricted maximum likelihood estimation, together with Bayesian estimation under corresponding unrestricted and order-restricted parameter spaces. The procedures are compared in a simulation study, which also examines the coverage of the interval estimators and the sensitivity of the Bayes estimators to the prior, and are illustrated on a real dataset.

The remainder of this paper is organized as follows. Section 2 describes the block Kumaraswamy model and the censoring scheme, constructs the likelihood, and presents unrestricted and order-restricted maximum likelihood estimation together with Fisher-information-based and per-block pivotal confidence intervals. Section 3 develops the Bayesian procedures with and without order restrictions, including posterior computation and credible intervals. Section 4 presents the simulation study and the real-data application. Section 5 concludes.

2 Model, Censoring Scheme and Maximum Likelihood Estimation

2.1 Block Kumaraswamy model

Consider k independent blocks of experimental units. We assume that the lifetime distribution in block i is Kumaraswamy with a common parameter α and a block-specific parameter λ_i. The parameter vector is

$$\boldsymbol{\theta} = (\alpha, \lambda_1, \ldots, \lambda_k), \qquad \alpha > 0,\ \lambda_i > 0. \tag{1}$$

For an observation in block i, the probability density function, cumulative distribution function and survival function are, for $0 < x < 1$,

$$f_i(x; \alpha, \lambda_i) = \alpha\lambda_i x^{\alpha-1}(1 - x^\alpha)^{\lambda_i - 1}, \tag{2}$$

$$F_i(x; \alpha, \lambda_i) = 1 - (1 - x^\alpha)^{\lambda_i}, \tag{3}$$

$$S_i(x; \alpha, \lambda_i) = (1 - x^\alpha)^{\lambda_i}. \tag{4}$$

Remark 1. Since $S_i(x) = (1-x^\alpha)^{\lambda_i}$ and $0 < 1-x^\alpha < 1$, the survival function is decreasing in λ_i. A larger λ_i therefore corresponds to stochastically shorter lifetimes. Consequently, the ordering $\lambda_1 \le \cdots \le \lambda_k$ considered later means that block 1 has the stochastically longest lifetimes and block k the shortest, so the blocks must be labelled accordingly. This interpretation requires that all blocks are measured on one common scale: when lifetimes are mapped into $(0,1)$, the same fixed increasing transformation must be used in every block. A block-specific rescaling (for example, dividing each block by its own maximum) removes the between-block differences that the λ_i are meant to capture, and the ordering of the λ_i then has no stochastic-ordering interpretation.

2.2 Block generalized adaptive progressive hybrid censoring

In block i, n_i units are placed on test, and the following quantities are fixed in advance: a target number of failures $r_i \le n_i$, two time thresholds $0 < T_{1i} < T_{2i} < \infty$, and a planned removal scheme $(R_{i1}, \ldots, R_{ir_i})$ with $\sum_{j=1}^{r_i} R_{ij} + r_i = n_i$. Let d_{1i} and d_{2i} denote the numbers of failures observed up to times T_{1i} and T_{2i}, respectively. Immediately after the j-th failure, R_{ij} surviving units are removed at random, with the following adaptation rule, which follows the generalized adaptive progressive hybrid censoring scheme of Lee and Lee [1] applied independently in each block:

- **Case I** ($X_{i,r_i} < T_{1i}$): the test stops at the r_i-th failure with the planned removals.
- **Case II** ($T_{1i} < X_{i,r_i} < T_{2i}$): after time T_{1i} no further units are removed, i.e. $R_{i,d_{1i}+1} = \cdots = R_{i,r_i-1} = 0$, and the test continues until the r_i-th failure, at which all remaining units are removed, so that $R_{i,r_i} = n_i - r_i - \sum_{j=1}^{d_{1i}} R_{ij}$.
- **Case III** ($X_{i,r_i} > T_{2i}$): after T_{1i} no further units are removed, and the test is terminated at time T_{2i} with d_{2i} observed failures; the $n_i - d_{2i} - \sum_{j=1}^{d_{1i}} R_{ij}$ units still on test are censored at T_{2i}.

The data from block i therefore consist of the m_i ordered failure times

$$x_{i1} < x_{i2} < \cdots < x_{im_i}, \tag{5}$$

the realized removals R_{ij} $(j = 1, \ldots, m_i)$, a terminal time τ_i, and a number C_i of units censored at τ_i, where the number of observed failures m_i is random in general. The realized quantities in each case are:

	Case I	Case II	Case III
m_i	r_i	r_i	d_{2i}
τ_i	x_{i,r_i}	x_{i,r_i}	T_{2i}
C_i	0	0	$n_i - d_{2i} - \sum_{j \le d_{1i}} R_{ij}$

In every case the unit-conservation identity

$$n_i = m_i + \sum_{j=1}^{m_i} R_{ij} + C_i \tag{6}$$

holds, where in Cases I and II the final removal R_{i,m_i} is part of the realized scheme and $C_i = 0$. We assume $m_i \ge 1$ in every block so that the quantities below are well defined (the pivotal intervals of Section 2.7 require $m_i \ge 2$).

Remark 2. The threshold T_{1i} affects the data only through the removals that are skipped after T_{1i}. If the planned scheme is of Type-II style, i.e. $R_{ij} = 0$ for $j < r_i$ and $R_{i,r_i} = n_i - r_i$, then there is nothing to skip: Cases I and II coincide and the design reduces to a Type-II censored sample truncated at T_{2i}. The adaptive feature is exercised only when the planned scheme removes units before the r_i-th failure.

2.3 Likelihood

An observed failure at x_{ij} contributes $f_i(x_{ij})$, the R_{ij} units removed alive at that time contribute $S_i(x_{ij})^{R_{ij}}$, and the C_i units remaining at the terminal time contribute $S_i(\tau_i)^{C_i}$. Because the removal decisions depend only on the history of the experiment (the number of failures before T_{1i} and the current failure time), the likelihood conditional on the realized scheme and for independent blocks is proportional to

$$L(\boldsymbol{\theta}) = \prod_{i=1}^{k} \left[\prod_{j=1}^{m_i} f_i(x_{ij};\alpha,\lambda_i)\, S_i(x_{ij};\alpha,\lambda_i)^{R_{ij}} \right] S_i(\tau_i;\alpha,\lambda_i)^{C_i}. \tag{7}$$

Substituting (2) and (4) gives

$$L(\boldsymbol{\theta}) = \prod_{i=1}^{k} \left[\prod_{j=1}^{m_i} \alpha\lambda_i x_{ij}^{\alpha-1}(1-x_{ij}^{\alpha})^{\lambda_i-1}(1-x_{ij}^{\alpha})^{\lambda_i R_{ij}} \right] (1-\tau_i^{\alpha})^{\lambda_i C_i}. \tag{8}$$

Taking logarithms, the log-likelihood is

$$\ell(\boldsymbol{\theta}) = \sum_{i=1}^{k} \Bigg[m_i \log\alpha + m_i \log\lambda_i + (\alpha-1)\sum_{j=1}^{m_i} \log x_{ij} + (\lambda_i - 1)\sum_{j=1}^{m_i} \log(1-x_{ij}^{\alpha}) + \lambda_i \sum_{j=1}^{m_i} R_{ij} \log(1-x_{ij}^{\alpha}) + C_i\lambda_i \log(1-\tau_i^{\alpha}) \Bigg]. \tag{9}$$

Define, for each block,

$$W_i(\alpha) = -\left[\sum_{j=1}^{m_i} (1+R_{ij}) \log(1-x_{ij}^{\alpha}) + C_i \log(1-\tau_i^{\alpha}) \right]. \tag{10}$$

Then the log-likelihood can be written as

$$\ell(\alpha,\boldsymbol{\lambda}) = \sum_{i=1}^{k} \left[m_i \log\alpha + m_i \log\lambda_i + (\alpha-1)\sum_{j=1}^{m_i} \log x_{ij} - \sum_{j=1}^{m_i} \log(1-x_{ij}^{\alpha}) - \lambda_i W_i(\alpha) \right], \tag{11}$$

where $\boldsymbol{\lambda} = (\lambda_1,\ldots,\lambda_k)$. Note the negative sign on the term $\sum_j \log(1-x_{ij}^{\alpha})$ in (11). The maximization is over $\Theta = \{(\alpha,\boldsymbol{\lambda}) : \alpha > 0, \lambda_i > 0\}$, and all expressions require $0 < x_{ij} < 1$ and $0 < \tau_i < 1$.

2.4 Unrestricted maximum likelihood estimation

The unrestricted maximum likelihood estimators (MLEs) maximize (11) over Θ without ordering constraints on the λ_i. For fixed α,

$$\frac{\partial \ell}{\partial \lambda_i} = \frac{m_i}{\lambda_i} - W_i(\alpha), \qquad \frac{\partial^2 \ell}{\partial \lambda_i^2} = -\frac{m_i}{\lambda_i^2} < 0, \tag{12}$$

so the conditional MLE of λ_i is the unique maximizer

$$\widehat{\lambda}_i(\alpha) = \frac{m_i}{W_i(\alpha)}, \qquad i = 1,\ldots,k, \tag{13}$$

provided $m_i > 0$ and $W_i(\alpha) > 0$. Substituting these into the log-likelihood gives the profile log-likelihood

$$\ell_p(\alpha) = \ell\big(\alpha, \widehat{\lambda}_1(\alpha),\ldots,\widehat{\lambda}_k(\alpha)\big), \tag{14}$$

and the MLE of the common parameter is

$$\widehat{\alpha} = \arg\max_{\alpha>0} \ell_p(\alpha). \tag{15}$$

Because α enters nonlinearly, $\widehat{\alpha}$ is obtained by numerical optimization over $\alpha > 0$; we evaluate ℓ_p on a fine grid in $\log \alpha$ and refine the best grid point by a one-dimensional search. Existence of an interior maximizer is not guaranteed for every sample (the profile log-likelihood can be monotone), so the search should be checked for solutions at the boundary of the search range; none occurred in the simulations or in the data analysis below. The unrestricted MLEs of the block-specific parameters are then

$$\widehat{\lambda}_i = \frac{m_i}{W_i(\widehat{\alpha})}, \qquad i = 1, \ldots, k. \tag{16}$$

2.5 Maximum likelihood estimation under order restrictions

When the scientific context suggests an ordering among the blocks, we consider

$$0 < \lambda_1 \le \lambda_2 \le \cdots \le \lambda_k, \qquad \alpha > 0, \tag{17}$$

interpreted as in Remark 1. For fixed α, the terms of ℓ that depend on $\boldsymbol{\lambda}$ are

$$Q(\boldsymbol{\lambda} \mid \alpha) = \sum_{i=1}^{k} \big[m_i \log \lambda_i - \lambda_i W_i(\alpha) \big], \tag{18}$$

which is jointly strictly concave in $\boldsymbol{\lambda}$ when all $m_i > 0$ and is maximized over a convex set (a polyhedral cone), so the constrained problem

$$\widehat{\lambda}^R(\alpha) = \arg\max_{\boldsymbol{\lambda}} Q(\boldsymbol{\lambda} \mid \alpha) \quad \text{subject to } 0 < \lambda_1 \le \cdots \le \lambda_k \tag{19}$$

has a unique maximizer. If the unrestricted conditional estimates $\widehat{\lambda}_i(\alpha)$ already satisfy the ordering, they solve (19). Otherwise adjacent blocks that violate the ordering are pooled. For a pooled group of consecutive blocks $i = r, \ldots, s$, maximizing $\sum_{i=r}^{s} [m_i \log \lambda - \lambda W_i(\alpha)]$ gives the common value

$$\widehat{\lambda}_{r:s}(\alpha) = \frac{\sum_{i=r}^{s} m_i}{\sum_{i=r}^{s} W_i(\alpha)}. \tag{20}$$

This pooling is carried out by the pool-adjacent-violators algorithm (PAVA) applied to the values $\widehat{\lambda}_i(\alpha)$ with weights $W_i(\alpha)$ (the pooled value is the W-weighted mean of the pooled $\widehat{\lambda}_i(\alpha)$). The restricted profile log-likelihood is

$$\ell_p^R(\alpha) = \ell\big(\alpha, \widehat{\lambda}_1^R(\alpha), \ldots, \widehat{\lambda}_k^R(\alpha)\big), \tag{21}$$

the order-restricted estimate of the common parameter is

$$\widehat{\alpha}^R = \arg\max_{\alpha>0} \ell_p^R(\alpha), \tag{22}$$

which again requires recomputing the conditional restricted estimates for each candidate α, and the restricted estimates of the block parameters are $\widehat{\lambda}_i^R = \widehat{\lambda}_i^R(\widehat{\alpha}^R)$, $i = 1, \ldots, k$. By construction they satisfy (17).

2.6 Fisher-information-based asymptotic confidence intervals

Let $\boldsymbol{\theta} = (\alpha, \lambda_1, \ldots, \lambda_k)^\top$. The observed Fisher information matrix is

$$J(\boldsymbol{\theta}) = -\frac{\partial^2 \ell(\boldsymbol{\theta})}{\partial \boldsymbol{\theta}\, \partial \boldsymbol{\theta}^\top}. \tag{23}$$

Evaluated at the unrestricted MLE $\widehat{\boldsymbol{\theta}}$, and under the usual regularity conditions, the covariance matrix of the estimator is approximated by $\widehat{\mathrm{Cov}}(\widehat{\boldsymbol{\theta}}) = J(\widehat{\boldsymbol{\theta}})^{-1}$, so that

$$\widehat{\mathrm{SE}}(\widehat{\theta}_j) = \sqrt{\big[J(\widehat{\boldsymbol{\theta}})^{-1}\big]_{jj}}. \tag{24}$$

An approximate $100(1-\gamma)\%$ confidence interval for θ_j is

$$\widehat{\theta}_j \pm z_{1-\gamma/2}\, \widehat{\mathrm{SE}}(\widehat{\theta}_j), \tag{25}$$

where $z_{1-\gamma/2}$ is the standard normal quantile ($z_{0.975} \approx 1.96$ for a 95% interval). Because every component of $\boldsymbol{\theta}$ is positive, the symmetric interval can have a negative lower limit in small samples; a delta-method interval on the log scale,

$$\Big[\widehat{\theta}_j \exp\big(-z_{1-\gamma/2}\widehat{\mathrm{SE}}(\widehat{\theta}_j)/\widehat{\theta}_j\big),\ \widehat{\theta}_j \exp\big(z_{1-\gamma/2}\widehat{\mathrm{SE}}(\widehat{\theta}_j)/\widehat{\theta}_j\big)\Big], \tag{26}$$

always lies in $(0, \infty)$ and is reported alongside it. These intervals apply to the unrestricted MLEs only; their accuracy depends on the large-sample approximation, and the effect of censoring and of the order constraint should be kept in mind when interpreting them.

2.7 Pivotal-based inference (working assumption)

Working assumption. As a working assumption, taken because it is the approach used in the reference paper, we adopt the pivotal-quantity method of Wu, Tseng and Kus [8] for the Kumaraswamy distribution under progressive Type-II censoring. Their results are stated for a single sample with parameters (λ, θ) and density $f(x; \lambda, \theta) = \lambda\theta x^{\lambda-1}(1-x^\lambda)^{\theta-1}$, which coincides with our $f_i(x; \alpha, \lambda_i)$ of (2) under the correspondence $\lambda \leftrightarrow \alpha$, $\theta \leftrightarrow \lambda_i$. We apply their pivots block by block, treating each block's realized censoring as if it were an ordinary pre-specified progressive Type-II scheme with m_i failures and removals $\tilde{R}_{ij}$, where $\tilde{R}_{ij} = R_{ij}$ for $j < m_i$ and $\tilde{R}_{i,m_i} = R_{i,m_i} + C_i$, so that any units censored at the terminal time are treated as removed at the last observed failure. In Cases I and II ($C_i = 0$) this is exactly the realized scheme; in Case III ($C_i > 0$, random m_i) it is an approximation, since the censored units are not removed at x_{i,m_i} but survive to T_{2i}. No joint pivot across several blocks is given in [8], so what follows is applied within each block separately, with n_i in place of their n.

Pivot for λ_i given α. Fix block i and α, and define $Y_{ij} = -\lambda_i \log(1 - x_{ij}^\alpha)$, $j = 1, \ldots, m_i$. By [8, Sect. 3.2], for a pre-specified progressive Type-II scheme the Y_{ij} form a progressively Type-II censored sample from a standard exponential distribution, and

$$V_i(\lambda_i, \alpha) = 2\sum_{j=1}^{m_i}(1 + \tilde{R}_{ij})Y_{ij} = 2\lambda_i \widetilde{W}_i(\alpha), \qquad \widetilde{W}_i(\alpha) = -\sum_{j=1}^{m_i}(1 + \tilde{R}_{ij})\log(1 - x_{ij}^\alpha), \tag{27}$$

follows a chi-square distribution with $2m_i$ degrees of freedom; this reproduces the result of [8, Eq. (8)] with their W. When $C_i = 0$, $\widetilde{W}_i(\alpha) = W_i(\alpha)$ of (10). A $100(1-\gamma)\%$ confidence interval for λ_i, for known α, follows directly from (27):

$$\left[\frac{\chi^2_{2m_i,1-\gamma/2}}{2\widetilde{W}_i(\alpha)},\ \frac{\chi^2_{2m_i,\gamma/2}}{2\widetilde{W}_i(\alpha)}\right], \tag{28}$$

where $\chi^2_{\nu,p}$ denotes the point with upper-tail probability p for the chi-square distribution with ν degrees of freedom. Since α is unknown in practice, (28) is useful mainly as a building block of the joint region below; plugging in $\widehat{\alpha}$ ignores the uncertainty in α.

Confidence interval for α, per block. Applying [8, Theorem 2] within block i (with $m_i \geq 2$), define

$$T_i(\alpha) = \frac{1}{n_i(m_i-1)} \sum_{j=1}^{m_i} (1+\tilde{R}_{ij}) \frac{\log(1-x_{ij}^{\alpha})}{\log(1-x_{i1}^{\alpha})} - \frac{1}{m_i-1}, \tag{29}$$

which, for a pre-specified progressive Type-II scheme, has an F distribution with $2m_i-2$ and 2 degrees of freedom and is a strictly monotone function of α on $(0,\infty)$ ([8, Lemma 4], under the correspondence above). Let $\varphi_i(\mathbf{x}_i, t)$ denote the unique solution in α of $T_i(\alpha) = t$. The $100(1-\gamma)\%$ confidence interval for α from block i alone is the interval whose endpoints are the two solutions $\varphi_i\big(\mathbf{x}_i, F_{1-\gamma/2}(2m_i-2,2)\big)$ and $\varphi_i\big(\mathbf{x}_i, F_{\gamma/2}(2m_i-2,2)\big)$, taken in increasing order, where $F_p(d_1,d_2)$ is the upper-p quantile of the F distribution with d_1, d_2 degrees of freedom (which endpoint is the lower one depends on whether T_i is increasing or decreasing in α). A joint confidence region for (α, λ_i) from block i follows by combining (29) with the independent pivot $V_i(\lambda_i, \alpha)$ of (27), as in [8, Theorem 3]. [8, Theorems 4–5] give a family of such intervals and regions indexed by $j = 2, \ldots, m_i$, obtained from generalized spacings anchored at the $(j-1)$-th order statistic rather than the first, with j chosen in practice to minimize interval length or region area; we do not reproduce the general $T_{ij}(\alpha)$ here but note that it is available if a tighter interval is wanted.

Remark 3. The exactness statements above hold for a pre-specified progressive Type-II scheme. Under the block GAPHC scheme they should be regarded as approximations, for three reasons. (i) The removal pattern is adaptive, so the realized scheme is random. (ii) Whether block i falls in Cases I/II or in Case III is decided by the event $\{X_{i,r_i} < T_{2i}\}$, which depends on the same failure times that enter the pivots; conditionally on the case, the pivots need not have exactly the χ^2 and F distributions. (iii) Case III, with its random m_i and censoring at T_{2i}, is outside the scope of [8]. Moreover, each interval uses only its own block's data and ignores the common α across blocks, so it is not the most efficient use of the sample, and no joint multi-block pivot combining all k blocks is given in [8] or derived here. The coverage of these intervals, overall and conditionally on the case, is therefore examined by simulation in Section 4.1.

3 Bayesian Estimation

Unlike the maximum likelihood approach, Bayesian estimation incorporates prior information about the unknown parameters through prior distributions. The observed data are combined with the prior via Bayes' theorem to obtain the posterior distribution. Let $\boldsymbol{\theta} = (\alpha, \lambda_1, \ldots, \lambda_k)$ and let $\mathbf{x}$ denote the observed data. Based on the likelihood $L(\boldsymbol{\theta})$ in (8), the joint posterior density is

$$\pi(\boldsymbol{\theta} \mid \mathbf{x}) = \frac{L(\boldsymbol{\theta})\pi(\boldsymbol{\theta})}{\int_{\Theta} L(\boldsymbol{\theta})\pi(\boldsymbol{\theta})\, d\boldsymbol{\theta}} \;\propto\; L(\boldsymbol{\theta})\pi(\boldsymbol{\theta}), \tag{30}$$

where $\pi(\boldsymbol{\theta})$ is the joint prior density.

3.1 Prior distributions

As a working assumption, we assign independent Gamma priors to the positive parameters, using the shape–rate parameterization:

$$\alpha \sim \text{Gamma}(a_0, b_0), \qquad \pi(\alpha) = \frac{b_0^{a_0}}{\Gamma(a_0)} \alpha^{a_0-1} e^{-b_0\alpha}, \quad \alpha > 0, \tag{31}$$

$$\lambda_i \sim \text{Gamma}(a_i, b_i), \qquad \pi(\lambda_i) = \frac{b_i^{a_i}}{\Gamma(a_i)} \lambda_i^{a_i-1} e^{-b_i\lambda_i}, \quad \lambda_i > 0, \tag{32}$$

for $i = 1, \ldots, k$, with all hyperparameters positive. The joint prior density is then

$$\pi(\boldsymbol{\theta}) = \pi(\alpha) \prod_{i=1}^{k} \pi(\lambda_i). \tag{33}$$

Independence of the components is an assumption made for tractability; other joint (dependent) priors can be substituted in (30). The order-restricted analysis of Section 3.3 uses the same prior truncated to the ordered set, which is itself a joint (dependent) prior on $(\lambda_1, \ldots, \lambda_k)$. Since the data enter the λ_i-conditionals through the quantities $W_i(\alpha)$, whose size depends on how the lifetimes were mapped into $(0, 1)$, the hyperparameters b_i should be chosen relative to that scale; Section 4.2 shows that this matters in practice.

Under squared-error loss the Bayes estimator of a parameter is its posterior mean:

$$\widehat{\alpha}_B = E[\alpha \mid \mathbf{x}], \qquad \widehat{\lambda}_{i,B} = E[\lambda_i \mid \mathbf{x}], \quad i = 1, \ldots, k. \tag{34}$$

3.2 Bayesian estimation without order restriction

Using the notation of Section 2, the likelihood can be written as

$$L(\boldsymbol{\theta}) = \alpha^{\sum_i m_i} \prod_{i=1}^{k} \lambda_i^{m_i} \prod_{i=1}^{k} \prod_{j=1}^{m_i} x_{ij}^{\alpha-1} \prod_{i=1}^{k} \prod_{j=1}^{m_i} (1 - x_{ij}^{\alpha})^{\lambda_i(1+R_{ij})-1} \prod_{i=1}^{k} (1 - \tau_i^{\alpha})^{\lambda_i C_i}. \tag{35}$$

Combining (35) with (33), and omitting constants that do not depend on $\boldsymbol{\theta}$, the joint posterior over the unrestricted space $\alpha > 0, \lambda_i > 0$ is

$$\begin{aligned} \pi(\boldsymbol{\theta} \mid \mathbf{x}) \propto \alpha^{a_0 + \sum_i m_i - 1} e^{-b_0 \alpha} \prod_{i=1}^{k} \lambda_i^{a_i + m_i - 1} e^{-b_i \lambda_i} \\ \times \prod_{i=1}^{k} \prod_{j=1}^{m_i} x_{ij}^{\alpha-1} (1 - x_{ij}^{\alpha})^{\lambda_i(1+R_{ij})-1} \prod_{i=1}^{k} (1 - \tau_i^{\alpha})^{\lambda_i C_i}. \end{aligned} \tag{36}$$

The Bayes estimators $\widehat{\alpha}_B$ and $\widehat{\lambda}_{i,B}$ in (34) are the means of the marginal posteriors, obtained by integrating (36) over the remaining parameters. Because of the nonlinear terms x_{ij}^{α} and τ_i^{α}, these expectations have no closed form, so simulation-based methods are used (Section 3.4).

3.3 Bayesian estimation under order restrictions

To incorporate the ordering (17), let

$$\Theta_R = \{(\alpha, \lambda_1, \ldots, \lambda_k) : \alpha > 0,\ 0 < \lambda_1 \leq \cdots \leq \lambda_k\}. \tag{37}$$

The likelihood is unchanged, and the restricted posterior is

$$\pi_R(\boldsymbol{\theta} \mid \mathbf{x}) = \frac{L(\boldsymbol{\theta})\pi(\boldsymbol{\theta})\, \mathbb{I}_{\Theta_R}(\boldsymbol{\theta})}{\int_{\Theta_R} L(\boldsymbol{\theta})\pi(\boldsymbol{\theta})\, d\boldsymbol{\theta}} \ \propto\ L(\boldsymbol{\theta})\pi(\boldsymbol{\theta})\, \mathbb{I}_{\Theta_R}(\boldsymbol{\theta}), \tag{38}$$

which is zero outside Θ_R. Equivalently, π_R is the unrestricted posterior (36) conditioned on the event Θ_R, whose posterior probability $P(\Theta_R \mid \mathbf{x})$ is itself informative about how strongly the data support the ordering. Under squared-error loss the order-restricted Bayes estimators are the means under (38):

$$\widehat{\alpha}_B^R = E_R[\alpha \mid \mathbf{x}], \qquad \widehat{\lambda}_{i,B}^R = E_R[\lambda_i \mid \mathbf{x}], \quad i = 1, \ldots, k. \tag{39}$$

Since the restricted posterior assigns probability only to ordered values, $\widehat{\lambda}_{1,B}^R \leq \widehat{\lambda}_{2,B}^R \leq \cdots \leq \widehat{\lambda}_{k,B}^R$.

3.4 Posterior computation

From (36), the full conditional distribution of λ_i given α and the data is available in closed form: collecting the terms in λ_i and using (10),

$$\lambda_i \mid \alpha, \mathbf{x} \;\sim\; \text{Gamma}\big(a_i + m_i,\; b_i + W_i(\alpha)\big), \quad i = 1, \ldots, k, \tag{40}$$

independently across i. Integrating the λ_i out of (36) gives the marginal posterior of α,

$$\pi(\alpha \mid \mathbf{x}) \propto \alpha^{a_0+M-1} e^{-b_0\alpha} \prod_{i,j} x_{ij}^{\alpha-1}(1-x_{ij}^{\alpha})^{-1} \prod_{i=1}^{k} \big[b_i + W_i(\alpha)\big]^{-(a_i+m_i)}, \qquad M = \sum_{i=1}^{k} m_i, \tag{41}$$

which is not of standard form but is a one-dimensional density. We therefore use a collapsed sampler: draw α from (41), then draw each λ_i from (40) given the drawn α. Draws from $\pi(\alpha \mid \mathbf{x})$ are obtained by evaluating (41) on a fine grid in $\log\alpha$ (4000 points on $[0.02, 30]$), normalizing, and sampling grid points with probability proportional to the density (a Metropolis–Hastings random walk on $\log\alpha$ targeting (41) is an alternative). This produces independent draws from the joint posterior, so no burn-in or autocorrelation diagnostics are needed; the only numerical error is the grid resolution, and the posterior mass near the ends of the grid should be checked to be negligible (it was negligible in all analyses reported here). Because π_R is the posterior conditioned on Θ_R, draws from the order-restricted posterior are obtained by retaining those joint draws that satisfy $\lambda_1 \le \cdots \le \lambda_k$; the retained fraction estimates $P(\Theta_R \mid \mathbf{x})$. When this fraction is very small, a truncated Gibbs sampler, in which the full conditional of λ_i is the Gamma distribution in (40) truncated to $[\lambda_{i-1}, \lambda_{i+1}]$ with $\lambda_0 = 0$ and $\lambda_{k+1} = \infty$, combined with a Metropolis–Hastings update of α, can be used instead. The Bayes estimates are the sample means of the draws.

3.5 Bayesian credible intervals

Posterior samples also provide interval estimates. For a parameter θ_j, a $100(1-\gamma)\%$ highest posterior density (HPD) credible interval is obtained from the ordered draws as the shortest interval containing a proportion $1-\gamma$ of the sample. The same construction applied to draws from (38) gives order-restricted credible intervals.

4 Numerical Illustration

This section evaluates the unrestricted and order-restricted MLE and Bayes procedures of Sections 2–3 by simulation, and applies them to a real dataset.The complete implementation code and simulation scripts used for testing and verifying these procedures are publicly available at `https://github.com/tankasalamadhughnesh-create/Pivotal-based-Classical-and-Bayesian-estimatio`
All results use $B = 1{,}000$ Monte Carlo replications for each setting; each Bayesian replication uses 20,000 independent posterior draws from the collapsed sampler of Section 3.4. Monte Carlo standard errors are given where differences between procedures are discussed.

4.1 Simulation study

We consider $k = 2$ blocks under the block GAPHC scheme of Section 2.2, with $n_1 = n_2 = 60$ and target failures $r_1 = r_2 = 40$. The planned removal scheme removes one surviving unit at each of the first 10 failures and the remaining 10 units at the 40th failure, i.e. $R_{ij} = 1$ for $j \le 10$, $R_{ij} = 0$ for $11 \le j \le 39$ and $R_{i,40} = 10$. This scheme has non-trivial removals before the r_i-th failure, so the threshold T_{1i} matters (Remark 2): failures occurring after T_{1i} no longer trigger removals. Data are generated sequentially under the full adaptation rule of Section 2.2, using the representation of a progressively censored sample through independent standard exponential

spacings, with the removal decisions made as failures occur and with the test stopped at T_{2i} when the next failure would exceed it. Two parameter configurations are considered, both with $\alpha = 2$:

- **Setting A (well separated):** $\lambda = (2.0, 4.0)$, $T_1 = (0.20, 0.15)$, $T_2 = (0.60, 0.50)$.
- **Setting B (close values):** $\lambda = (2.5, 3.0)$, $T_1 = (0.20, 0.20)$, $T_2 = (0.60, 0.60)$.

The realized cases were as follows. Case I never occurred, because the 40th failure essentially never precedes T_{1i} at these thresholds. In Setting A, block 1 realized Case II in 2.9% and Case III in 97.1% of replications, and block 2 in 29.7% and 70.3%; in Setting B the corresponding figures are 20.2%/79.8% and 52.5%/47.5%. The average numbers of observed failures were (32.9, 37.2) in Setting A and (36.4, 38.5) in Setting B. No replication produced $m_i < 2$ in either block, and the profile search never returned a boundary solution.

Maximum likelihood estimation. Setting B is included specifically to exercise the PAVA pooling step of Section 2.5, since a small gap between λ_1 and λ_2 makes the unrestricted conditional estimates more likely to violate the ordering $\lambda_1 \leq \lambda_2$ in a given replication. Pooling occurred in 21.1% of the replications in Setting B and in only 0.2% in Setting A. Table 1 reports the bias and MSE of the unrestricted and restricted MLEs together with the paired difference in squared error (restricted minus unrestricted) and its Monte Carlo standard error.

Table 1: Bias and MSE of the unrestricted and restricted MLEs ($n_i = 60$, $r_i = 40$, $B = 1{,}000$). The last column is the mean paired difference in squared error (restricted − unrestricted) with its Monte Carlo standard error in parentheses. Negative values favour the restricted estimator.

		Unrestricted MLE		Restricted MLE		
Setting	Parameter	Bias	MSE	Bias	MSE	ΔMSE (SE)
A	α	0.048	0.066	0.048	0.066	0.0000 (0.0000)
	λ_1	0.142	0.346	0.142	0.345	−0.0004 (0.0003)
	λ_2	0.375	2.079	0.376	2.079	−0.0001 (0.0001)
B	α	0.058	0.067	0.057	0.067	−0.0003 (0.0001)
	λ_1	0.208	0.566	0.159	0.486	−0.0797 (0.0123)
	λ_2	0.280	0.933	0.316	0.927	−0.0059 (0.0042)

In Setting A the two estimators are practically identical, since pooling almost never occurs. In Setting B the restriction reduces the MSE of λ_1 from 0.566 to 0.486 (a 14% reduction; paired difference −0.080 with standard error 0.012) and moves some of its upward bias (0.208 versus 0.159) to λ_2 (bias 0.280 versus 0.316). The change in the MSE of λ_2 (0.933 versus 0.927) is within Monte Carlo error (paired difference −0.006, standard error 0.004), and the effect on α is negligible. The MLEs of the λ_i are biased upwards at these sample sizes in both settings.

Interval estimation. Table 2 reports the empirical coverage of nominal 95% intervals. The Fisher-information intervals, in both the normal and the log-scale form, and the HPD credible intervals (with Gamma$(2, 1)$ priors) have coverage within about 0.02 of nominal. The Monte Carlo standard error of each entry is at most 0.007. The pivotal row combines the per-block intervals of Section 2.7: the α column pools the intervals from both blocks, and the λ_i columns use the interval (28) evaluated at the true α.

Table 3 splits the pivotal coverage by the case realized in the block, as discussed in Remark 3. Pooled over cases the coverage is close to nominal, but conditionally on Case II the intervals are somewhat too short (coverage 0.926–0.943 for α and as low as 0.724 for λ in the 29 replications of

Table 2: Empirical coverage of nominal 95% interval estimators ($B = 1{,}000$).

	Setting A			Setting B		
Method	α	λ_1	λ_2	α	λ_1	λ_2
Fisher, normal	0.960	0.953	0.956	0.946	0.959	0.949
Fisher, log-scale	0.941	0.947	0.954	0.936	0.943	0.942
Bayes HPD, Gamma$(2, 1)$	0.964	0.942	0.932	0.962	0.957	0.946
Pivotal (per block, pooled over cases)	0.947	0.954	0.960	0.953	0.959	0.956

Setting A, block 1, whose standard error is about 0.08), while conditionally on Case III they are slightly conservative (0.947–0.966 for α). This is consistent with the selection effect described in Remark 3(ii) and is why we describe the intervals as approximate under the adaptive scheme.

Table 3: Coverage of the per-block pivotal 95% intervals by realized case; λ_i intervals use the true α. "Reps" is the number of replications in which the block realized that case.

Setting	Block	Case	Reps	α	λ_i
A	1	II	29	0.931	0.724
	1	III	971	0.953	0.961
	2	II	297	0.926	0.923
	2	III	703	0.947	0.976
B	1	II	202	0.936	0.926
	1	III	798	0.956	0.967
	2	II	525	0.943	0.956
	2	III	475	0.966	0.956

Bayesian estimation. Table 4 reports the Bayes estimates (posterior means under squared-error loss) with independent Gamma$(2, 1)$ priors for α and each λ_i, whose mean 2 equals the true value of α. The restricted columns use the replications in which at least 500 of the 20,000 draws satisfied the ordering (997 of 1,000 in Setting B and all 1,000 in Setting A); the posterior probability of the ordering averaged 0.97 in Setting A and 0.71 in Setting B.

Table 4: Bias and MSE of the unrestricted and restricted Bayes estimates (posterior means), Gamma$(2, 1)$ priors, $B = 1{,}000$.

		Unrestricted Bayes		Restricted Bayes	
Setting	Parameter	Bias	MSE	Bias	MSE
A	α	-0.067	0.038	-0.066	0.038
	λ_1	-0.034	0.171	-0.045	0.165
	λ_2	-0.289	0.686	-0.268	0.654
B	α	-0.027	0.039	-0.027	0.039
	λ_1	-0.003	0.266	-0.155	0.217
	λ_2	-0.029	0.384	0.123	0.365

With this prior the Bayes estimators have smaller MSE than the MLEs of Table 1, for example 0.039 versus 0.067 for α and 0.266 versus 0.566 for λ_1 in Setting B, at the price of a small bias towards the prior mean. The order restriction lowers the MSE of both λ_1 ($0.266 \to 0.217$) and λ_2 ($0.384 \to 0.365$) in Setting B and changes little in Setting A, where the ordering is almost

certain a posteriori. The restriction shifts the posterior means of the two λ_i apart, producing a negative bias for λ_1 and a positive bias for λ_2 in Setting B.

Prior sensitivity. Table 5 repeats the unrestricted Bayes analysis under five prior choices (the same Gamma(a, b) prior is used for α and each λ_i). The prior mean a/b equals 2, 1, 1, 2 and 4, respectively, with prior variances $a/b^2 = 2$, 1, 10, 1 and 2. The estimates move substantially with the prior: the bias of α in Setting B ranges from -0.121 (mean 1) to $+0.235$ (mean 4), and that of λ_2 from -0.304 to $+0.759$. A prior whose mean is well above the truth, Gamma$(8, 2)$, gives larger MSE than the MLE for every parameter in Setting B, whereas the nearly noninformative Gamma$(0.1, 0.1)$ prior gives smaller MSEs than the MLE for every parameter in both settings (for example 0.500 versus 0.566 for λ_1 and 0.813 versus 0.933 for λ_2 in Setting B). With roughly 40 failures per block the likelihood therefore does not dominate informative priors, and prior choice should be reported and examined.

Table 5: Bias and MSE of the unrestricted Bayes estimates under different Gamma(a, b) priors ($B = 1{,}000$).

		α		λ_1		λ_2	
Setting	Prior	Bias	MSE	Bias	MSE	Bias	MSE
A	Gamma$(2, 1)$	-0.067	0.038	-0.034	0.171	-0.289	0.686
	Gamma$(1, 1)$	-0.166	0.059	-0.220	0.192	-0.681	0.955
	Gamma$(0.1, 0.1)$	0.009	0.057	0.097	0.302	0.259	1.697
	Gamma$(4, 2)$	-0.118	0.037	-0.109	0.132	-0.611	0.694
	Gamma$(8, 2)$	0.188	0.064	0.556	0.502	0.685	0.994
B	Gamma$(2, 1)$	-0.027	0.039	-0.003	0.266	-0.029	0.384
	Gamma$(1, 1)$	-0.121	0.050	-0.230	0.274	-0.304	0.403
	Gamma$(0.1, 0.1)$	0.027	0.059	0.161	0.500	0.219	0.813
	Gamma$(4, 2)$	-0.068	0.033	-0.118	0.190	-0.202	0.276
	Gamma$(8, 2)$	0.235	0.090	0.694	0.776	0.759	0.989

4.2 Real-data application

We use breakdown-time data for an electrical insulating fluid tested at several voltage stress levels (Nelson [6]; the full dataset is reproduced in Ramsey and Schafer [7], Dataset 0802). Times to breakdown (in minutes) were recorded at seven voltage levels; we use the four levels with the largest sample sizes, 30, 32, 34 and 36 kV, giving $k = 4$ blocks of sizes $n = (11, 15, 19, 15)$. Higher voltage produces earlier breakdown, so we expect the blocks, ordered by increasing voltage, to have stochastically decreasing lifetimes, i.e. increasing λ_i by Remark 1.

Data preparation. The Kumaraswamy distribution requires support $(0, 1)$, while the raw breakdown times range from under a minute to a few hundred minutes. Following Remark 1, we map the times into $(0, 1)$ with one common transformation for all blocks, $x = t/250$ with t in minutes; the divisor is a fixed constant larger than every observed time and is not computed from the data of any block. This preserves the between-block differences in lifetime, so that the ordering of the λ_i retains its meaning. The magnitudes of the λ_i depend on the divisor (for divisors 250, 500 and 1000 the MLE of α is 0.806, 0.820 and 0.826, and the $\widehat{\lambda}_i$ roughly double with each doubling of the divisor, e.g. $\widehat{\lambda}_1 = 1.84, 3.72, 7.06$, while their ordering is unchanged), so only the ordering and the ratios of the λ_i should be interpreted, not their absolute values. Because the data are fully observed, we impose the block GAPHC scheme of Section 2.2 ourselves, for illustration: in each block we take $r_i = \lceil 0.7 n_i \rceil$, a Type-II-style planned removal scheme, and

a common time threshold $T_{2i} = 100$ minutes (chosen for illustration). With a Type-II-style plan the threshold T_{1i} plays no role (Remark 2), so the analysis exercises the terminal-censoring branch (Case III) but not the adaptive removals. The 30 kV block realized Case III ($m = 7$ failures by 100 minutes and $C = 4$ units censored at T_2) and the other three realized Case II ($m_i = r_i$). The numbers of observed failures are $m = (7, 11, 14, 11)$.

Maximum likelihood estimates. The unrestricted MLEs are

$$\widehat{\alpha} = 0.806, \qquad \widehat{\lambda} = (1.836,\ 4.850,\ 13.888,\ 31.908).$$

They already satisfy $\widehat{\lambda}_1 < \widehat{\lambda}_2 < \widehat{\lambda}_3 < \widehat{\lambda}_4$, so the order-restricted MLEs coincide with them (no pooling is required), and the increase of $\widehat{\lambda}_i$ with voltage matches the physical expectation of Remark 1. Table 6 gives the log-scale Fisher intervals (26); the symmetric normal interval for λ_4 has a negative lower limit (-7.1), which illustrates why the log-scale form is preferred. The intervals for the λ_i are wide and overlap, so the ordering is suggestive for adjacent voltages rather than established by the intervals alone.

Table 6: Unrestricted MLEs and 95% log-scale Fisher confidence intervals for the insulating-fluid data (the order-restricted MLEs are identical).

Parameter	MLE	95% interval
α	0.806	(0.604, 1.074)
λ_1 (30 kV)	1.836	(0.798, 4.223)
λ_2 (32 kV)	4.850	(2.232, 10.539)
λ_3 (34 kV)	13.888	(5.228, 36.889)
λ_4 (36 kV)	31.908	(9.383, 108.504)

Pivotal confidence intervals and the common-α assumption. Applying the per-block interval of Section 2.7 to each block separately gives the 95% intervals in Table 7. The 30 kV interval is approximate (Case III), and the others are subject to the caveats of Remark 3. The intervals overlap, which is consistent with a common α but does not establish it. A likelihood-ratio test of a common α against block-specific α_i (blockwise estimates $0.841, 0.453, 0.970, 1.508$) gives $2\Delta\ell = 9.26$ on 3 degrees of freedom (asymptotic $p = 0.026$), i.e. moderate evidence of heterogeneity in α across voltages, so the common-α model should be regarded as an approximation for these data. As a descriptive fit check, the Kolmogorov–Smirnov distance of the 43 normalized spacings of the fitted model from the standard exponential distribution is 0.053 (parametric-bootstrap $p = 0.99$ with 300 replicates), which shows no evidence of misfit, although the power of such a check with 43 spacings is limited.

Table 7: Per-block 95% pivotal confidence intervals for α (Theorem 2 of [8], applied as in Section 2.7) and blockwise MLEs of α. The 30 kV interval (Case III) is approximate.

Block (voltage)	n_i	m_i	Case	$\widehat{\alpha}_i$	95% interval for α
30 kV	11	7	III	0.841	(0.453, 3.356)
32 kV	15	11	II	0.453	(0.235, 1.293)
34 kV	19	14	II	0.970	(0.385, 1.694)
36 kV	15	11	II	1.508	(0.627, 2.881)

Bayesian estimates. We use independent nearly noninformative Gamma$(0.1, 0.1)$ priors for α and each λ_i and 10^6 independent posterior draws. The unrestricted and restricted posterior means and 95% HPD intervals (computed from at least 4×10^5 draws) are given in Table 8. The posterior probability of the ordering is $P(\Theta_R \mid \mathbf{x}) = 0.874$, with pairwise posterior probabilities $P(\lambda_1 \le \lambda_2) = 0.969$, $P(\lambda_2 \le \lambda_3) = 0.979$ and $P(\lambda_3 \le \lambda_4) = 0.923$. The data therefore support the physical ordering, and the restricted and unrestricted posteriors are close; imposing the restriction mainly moves the 36 kV mean up slightly and narrows its interval from below. The posterior means lie below the MLEs, the more so for the larger λ_i (compare Table 9).

Table 8: Bayes estimates (posterior mean, squared-error loss) and 95% HPD intervals for the insulating-fluid data, Gamma$(0.1, 0.1)$ priors.

	Unrestricted		Restricted	
Parameter	Mean	95% HPD	Mean	95% HPD
α	0.671	(0.501, 0.840)	0.674	(0.509, 0.846)
λ_1 (30 kV)	1.458	(0.44, 2.64)	1.436	(0.45, 2.57)
λ_2 (32 kV)	3.545	(1.31, 6.12)	3.581	(1.42, 6.08)
λ_3 (34 kV)	8.409	(2.77, 15.33)	8.366	(2.91, 14.99)
λ_4 (36 kV)	15.607	(4.00, 30.36)	16.314	(4.79, 31.00)

Table 9 shows how the unrestricted posterior means depend on the prior. The estimates of the λ_i are strongly prior-dependent: for λ_4 the posterior mean ranges from 4.7 (Gamma$(1, 1)$) to 23.3 (Gamma$(1, 0.1)$), against an MLE of 31.9. The reason is that $W_i(\alpha)$ is small on this scale (for example $W_4(\widehat{\alpha}) = m_4/\widehat{\lambda}_4 \approx 0.34$), so a prior rate $b_i = 1$ carries more weight in (40) than the data do; the prior rate should be set relative to the normalization of the lifetimes. The conclusion about ordering is more stable than the point estimates: $P(\Theta_R \mid \mathbf{x})$ rises from 0.64 to 0.91 as the prior is made weaker, and the posterior means increase with voltage under every prior.

Table 9: Sensitivity of the unrestricted posterior means (and of $P(\Theta_R \mid \mathbf{x})$) to the Gamma$(a, b)$ prior used for all parameters; the last row gives the MLEs.

Prior	α	λ_1	λ_2	λ_3	λ_4	$P(\Theta_R \mid \mathbf{x})$
Gamma$(1, 1)$	0.490	1.031	2.065	3.553	4.741	0.637
Gamma$(2, 1)$	0.535	1.254	2.462	4.314	5.820	0.670
Gamma$(0.1, 0.1)$	0.671	1.458	3.545	8.409	15.607	0.874
Gamma$(1, 0.1)$	0.756	1.896	4.613	11.880	23.269	0.915
MLE	0.806	1.836	4.850	13.888	31.908	—

5 Conclusion

This study developed classical and Bayesian estimation for the Kumaraswamy distribution under a block generalized adaptive progressive hybrid censoring scheme, with a common shape parameter α and block-specific parameters λ_i. We derived the likelihood under the adaptive scheme (Section 2.2), obtained unrestricted and order-restricted MLEs via profiling and pool-adjacent-violators pooling (Sections 2.4 –2.5), gave Fisher-information-based intervals (Section 2.6) and, as a working assumption drawn from Wu, Tseng and Kus [8], per-block pivotal intervals for α and λ_i (Section 2.7), and developed unrestricted and order-restricted Bayesian estimation with an exact collapsed posterior sampler (Sections 3.2–3.5).

In the simulation study (Section 4.1) the order restriction reduced the MSE of the lower block parameter when the true λ_i were close, with no significant change for the upper one and no effect on α, and it had practically no effect when the λ_i were well separated, since pooling then rarely occurs. The Fisher and HPD intervals had coverage close to nominal. The pivotal intervals had near-nominal coverage when pooled over cases, but they are not exact under the adaptive scheme and showed departures conditionally on the realized case. The Bayes estimators had smaller MSE than the MLEs under well-centred or nearly noninformative priors and larger MSE under a prior that was badly centred, so their performance depends materially on the prior. In the real-data application to the insulating-fluid breakdown times (Section 4.2), the MLEs of the λ_i increased with voltage as the physical reasoning predicts, so no pooling was needed, and the posterior probability of the ordering was 0.64–0.91 depending on the prior.

Several limitations should be kept in mind. First, the pivotal intervals of Section 2.7 are applied block by block and are approximate under GAPHC; no joint pivot combining all k blocks' information about the shared α is available from [8] or derived here, so these intervals are wider than a hypothetical combined interval would be. Second, the simulation covers $k = 2$ blocks and one family of planned schemes, and Case I did not occur. Third, the real-data illustration imposed the censoring scheme on complete data rather than analyzing data collected under that scheme; the time threshold and the normalizing divisor are illustrative choices, the λ_i are interpretable only up to that divisor, the Type-II-style plan leaves the adaptive removals unexercised, and a likelihood-ratio test gave moderate evidence ($p = 0.026$) against a common α. Fourth, the Bayesian estimates, especially of the λ_i, are sensitive to the prior, so the prior and its scale relative to the data normalization should always be reported together with a sensitivity analysis. Natural next steps are a joint multi-block pivot, studies with more than two blocks and with planned schemes that exercise Case I, dependent joint priors for the λ_i, and an analysis of data actually collected under a block GAPHC design.

References

[1] H. Lee and K. Lee, "Exact likelihood inference for an exponential parameter under generalized adaptive progressive hybrid censoring," *Symmetry*, vol. 12, no. 7, p. 1149, 2020.

[2] L. Wang, S.-J. Wu, H. Lin, and Y. M. Tripathi, "Inference for block progressive censored competing risks data from an inverted exponentiated exponential model," *Quality and Reliability Engineering International*, vol. 39, pp. 2736–2764, 2023.

[3] F. Sultana, C. Cetinkaya, M. Z. Raqab, and D. Kundu, "Inference for block type-I hybrid censored Weibull populations," *Communications in Statistics – Theory and Methods*, vol. 55, no. 11, pp. 3436–3459, 2026.

[4] A. K. Mahto, C. Lodhi, Y. M. Tripathi, and L. Wang, "Inference for partially observed competing risks model for Kumaraswamy distribution under generalized progressive hybrid censoring," *Journal of Applied Statistics*, vol. 49, no. 8, pp. 2064–2092, 2022.

[5] R. Kumari, Y. M. Tripathi, L. Wang, and R. K. Sinha, "Reliability estimation for Kumaraswamy distribution under block progressive Type-II censoring," *Statistics*, vol. 58, no. 1, pp. 142–175, 2024.

[6] W. B. Nelson, *Applied Life Data Analysis*. Wiley, New York, 1982.

[7] F. L. Ramsey and D. W. Schafer, *The Statistical Sleuth: A Course in Methods of Data Analysis*, 2nd ed. Duxbury, 2002. (Dataset case0802, distributed in the `Sleuth2` R package.)

[8] S.-J. Wu, Y.-T. Tseng, and C. Kus, "Likelihood and pivotal inference for Kumaraswamy parameters under progressive Type-II censoring," *The Journal of Supercomputing*, vol. 81, art. 1029, 2025.